

Question: Expected Threat (xT) Simulation

Consider a simplified football model with 3 zones:

- 0 → Central
- 1 → Wing
- 2 → Box

The transition matrix is:

$$A = \begin{bmatrix} 0.3 & 0.4 & 0.2 \\ 0.2 & 0.5 & 0.1 \\ 0.1 & 0.2 & 0.4 \end{bmatrix}$$

The goal probabilities are:

$$g = [0.05, 0.03, 0.20]$$

A possession starts from the **Central** zone.

The following random numbers are generated during the simulation:

$$r_1 = 0.65, \quad r_2 = 0.92$$

- (a) [4 marks] Trace the possession step-by-step using cumulative probabilities. Clearly show the state transitions.
- (b) [2 marks] What is the final outcome of the possession? (Goal or Out of play)

Solution

Step 1: From Central (State 0)

$$A_{\text{Central}} = [0.3, 0.4, 0.2]$$

Cumulative probabilities:

$$[0.3, 0.7, 0.9]$$

Using $r_1 = 0.65$:

$$0.3 < 0.65 < 0.7 \Rightarrow \text{move to Wing}$$

Central $\rightarrow$ Wing

Step 2: From Wing (State 1)

$$A_{\text{Wing}} = [0.2, 0.5, 0.1]$$

Cumulative probabilities:

$$[0.2, 0.7, 0.8]$$

Using $r_2 = 0.92$:

$$r_2 > 0.8 \Rightarrow \text{possession ends}$$

Now check if it is a goal:

$$g[1] = 0.03$$

Goal interval:

$$0.8 \rightarrow 0.83$$

Since:

$$r_2 = 0.92 > 0.83$$

$$\Rightarrow \text{Out of play}$$

Final Answer:

- Possession path: Central $\rightarrow$ Wing $\rightarrow$ Out of play
- Outcome: Not a goal